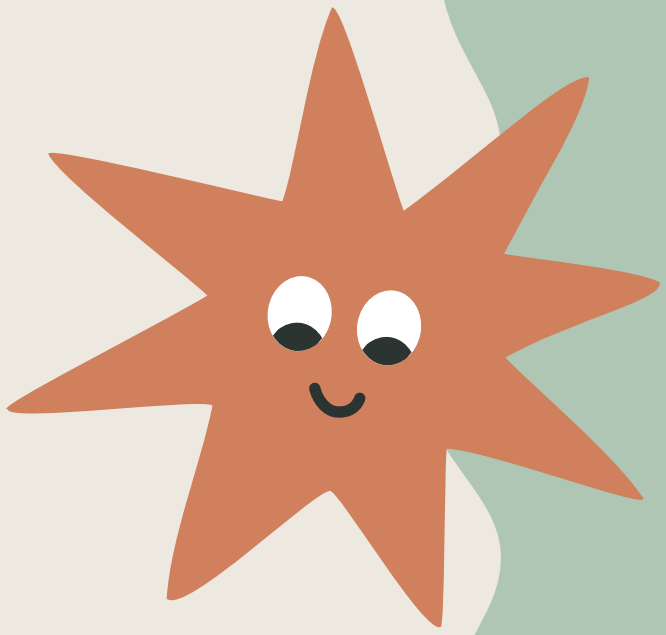
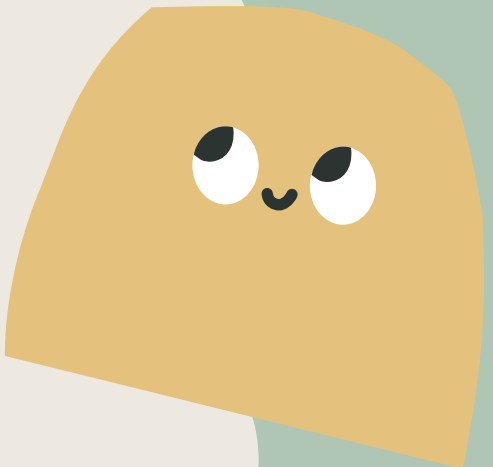
Variables & Data Types

Lecture 2





Learning Objectives

➢By the end of this lecture, students should be able to:

- Understand what a variable is.
- Create and assign variables in Python.
- Understand Python's dynamic typing.
- Identify common Python data types.
- Use int, float, str, and bool.
- Check a variable's type using type().
- Convert values between data types.
- Follow basic variable naming rules.
- Use variables in simple AI-related programs.

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What is a Variable?

- A variable is a name used to store a value in a program
- We don't need to mention data type

```
name = "Ali"
age = 20
marks = 85
```

- Python provides a convenient way to assign multiple variables

```
x, y, z = 10, 20, 30
```

```
x = y = z = 0
```

- Naming variables: Valid and invalid names
- Naming convention: snake_case

```
student_name = "Ali"
total_marks = 450
student_age = 20
```

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Main Python Data Types

Data Type	Python Type	Example
Integer	`int`	`20`
Floating point	`float`	`20.5`
String	`str`	`"Ali" or 'Ali'`
Boolean	`bool`	`True`

- There are more Python data types, and we will study lists, tuples, sets, and dictionaries in a later lecture
- Checking Data Type with type()
- Python is dynamically typed language

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Type Conversion

```
age = "20"

age = int(age)

print(age)
print(type(age))
```

```
x = 10

y = float(x)

print(y)
```

```
age = 20

age_text = str(age)

print(age_text)
print(type(age_text))
```

```
age = 20

print("My age is " + str(age))
```

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